MATH 3330 Homework 1

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1. Show that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Proof: We must show that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ by proving that $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$ and that $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$.

Let $x \in A \cap (B \cup C)$. Then it must be true that $x \in A$ and $x \in B \cup C$. So if $x \in B$, then $x \in A \cap B$. Similarly, if $x \in C$, then $x \in A \cap C$. Since $x \in B \cup C$ (that is, x is a member of B, or x is a member of C, or x is a member of both B and C), one or both of the two previous statements are true, so $x \in (A \cap B) \cup (A \cap C)$. Therefore, $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.

Let $x \in (A \cap B) \cup (A \cap C)$. Then $x \in A \cap B$ or $x \in A \cap C$. So, $x \in A$ and x must be a member of B or C. Thus, $x \in A \cap (B \cup C)$, so therefore $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$.

Since we have shown $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$ and $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$, it must be true that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

2. Show that $A - (B \cap C) = (A - B) \cup (A - C)$.

Proof: We must show that $A - (B \cap C) = (A - B) \cup (A - C)$ by proving that $A - (B \cap C) \subseteq (A - B) \cup (A - C)$ and that $(A - B) \cup (A - C) \subseteq A - (B \cap C)$.

Let $x \in A - (B \cap C)$. That means that $x \in A$, but $x \notin B \cap C$ (so $x \notin B$ or $x \notin C$). If $x \notin B$, then $x \in A - B$. Similarly, if $x \notin C$, then $x \in A - C$. Since one or both of the two previous statements must be true, $x \in (A - B) \cup (A - C)$. Therefore, $A - (B \cap C) = (A - B) \cup (A - C)$.

Let $x \in (A - B) \cup (A - C)$. Then $x \in A - B$ or $x \in A - C$. So, $x \in A$, and $x \notin B$ or $x \notin C$, implying that $x \notin B \cap C$. Thus, $x \in A - (B \cap C)$, and therefore $(A - B) \cup (A - C) \subseteq A - (B \cap C)$.

Since we have shown that $A - (B \cap C) \subseteq (A - B) \cup (A - C)$ and that $(A - B) \cup (A - C) \subseteq A - (B \cap C)$, it must be true that $A - (B \cap C) = (A - B) \cup (A - C)$.

3. True or False? $(A \cap B) \cup (A - B) = A$

This statement is true.

Proof: Assume that $A - B = A \cap B^c$. Then $(A \cap B) \cup (A - B) = (A \cap B) \cup (A \cap B^c)$. From **#1**, we've shown that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, letting $B^c = C$, we have $(A \cap B) \cup (A \cap B^c) = A \cap (B \cup B^c)$. Since $B \cup B^c = U$ (the universal set U), so $A \cap (B \cup B^c) = A \cap U = A$. Therefore, $(A \cap B) \cup (A - B) = A$ is true.

4. True or False? $(A \times B) \cup (C \times D) = (A \cup C) \times (B \cup D)$

This statement is false.

Counterexample: Take $A = \{1\}$, $B = \{1\}$, $A = \{2\}$, $D = \{2\}$. Then the left hand side would be $\{(1,1),(2,2)\}$, while the right hand side would be $\{(1,1),(1,2),(2,1),(2,2)\}$. Since the left hand and right hand sides are not equivalent, the statement is false.

What is true, however, is the subset $(A \times B) \cup (C \times D) \subseteq (A \cup C) \times (B \cup D)$.

5. Consider the following statement:

$$\text{If } x > 15 \text{ then } x^2 - 1 < 0$$

1. **Converse:** If $x^2 - 1 < 0$, then $x > 15$.
2. **Contrapositive:** If $x^2 - 1 \geq 0$, then $x \leq 15$.
3. **Inverse:** If $x \leq 15$, then $x^2 - 1 \geq 0$.
4. None of the above statements are true

6. What is the negation of the following statements?

5. There exists some $n \in \mathbb{Z}$ such that $n^2 + n$ is not composite.
6. $\forall a \in A$, $a \in B$.

7. Write $\mathbb{R} - \mathbb{Z}$ as a union of open intervals.

$$\mathbb{R} - \mathbb{Z} = \bigcup_{x \in \mathbb{Z}} (x, x+1)$$

8. Construct a collection of sets $A \in \mathcal{A}$ such that each A is an open interval in $\mathbb{R}$ and $\bigcap_{A \in \mathcal{A}} A$ is a single point.

We choose a point $x \in \mathbb{R}$ and define

$$\mathcal{A} = \{A_n \mid n \in \mathbb{N}\} \text{ where } A_n = (x - n, x + n) \text{ an interval on } \mathbb{R}$$

We claim that $\bigcap_{A \in \mathcal{A}} A = \{x\}$. Clearly $x \in A_n$ for all $n \in \mathbb{N}$. Then, consider when $n = 0$. In this case, $A_0 = (x - 0, x + 0) = (x, x) = \{x\}$. Since we are taking the intersection over all $n \in \mathbb{N}$ and there exists an element that we are intersecting that contains only one element, the resulting set must contain at most one element. Therefore, $\bigcap_{A \in \mathcal{A}} A = \{x\}$.

9 (optional). Formulate and prove DeMorgan's law for arbitrary unions and intersections.

Proof: Let X be a universal set and $\mathcal{A}$ be an arbitrary collection of subsets of X. We must prove both of the following statements:

Complement of a union:

$$\left(\bigcup_{A \in \mathcal{A}} A\right)^c = \bigcap_{A \in \mathcal{A}} A^c$$

Take any $x \in X$. Then,

$$\begin{aligned} x \in \left(\bigcup_{A \in \mathcal{A}} A\right)^c &\iff x \notin \bigcup_{A \in \mathcal{A}} A \\ &\iff \forall A \in \mathcal{A}, x \notin A \\ &\iff \forall A \in \mathcal{A}, x \in A^c \\ &\iff x \in \bigcap_{A \in \mathcal{A}} A^c \end{aligned}$$

Since this is true for all $x \in X$, the sets must be equal.

Complement of an intersection:

$$\left(\bigcap_{A \in \mathcal{A}} A\right)^c = \bigcup_{A \in \mathcal{A}} A^c$$

Take any $x \in X$. Then,

$$\begin{aligned} x \in \left(\bigcup_{A \in \mathcal{A}} A\right)^c &\iff x \notin \bigcup_{A \in \mathcal{A}} A \\ &\iff \exists A \in \mathcal{A}, x \notin A \\ &\iff \exists A \in \mathcal{A}, x \in A^c \\ &\iff x \in \bigcup_{A \in \mathcal{A}} A^c \end{aligned}$$

Since this is true for all $x \in X$, the sets must be equal.